

Computer Networks (CN)

Comprehensive Theory, Core Concepts, Real-World Examples & 25 Practice Questions

1. Introduction & Computer Networking Theory

A **Computer Network** is a system of interconnected digital devices (computers, servers, routers, switches) that exchange data and share resources using communication protocols over physical or wireless transmission media.

Key Objectives of Computer Networking

- **Resource Sharing:** Sharing hardware (printers, storage) and software resources.
- **Communication:** Fast exchange of data via email, instant messaging, and video streaming.
- **Reliability & Backup:** Alternative paths ensure high availability and data redundancy.
- **Scalability:** Easy addition of new devices without disrupting existing setups.

2. Core Concepts & Examples

1. Types of Networks (By Geographical Scale)

- **PAN (Personal Area Network):** Covers small range around a single person (~10 meters).
- **LAN (Local Area Network):** Connects devices within a single building or room.
- **MAN (Metropolitan Area Network):** Covers a town or city (~50 km).
- **WAN (Wide Area Network):** Spans countries or globally across continents.

Real-World Example: Connecting your smartphone to a Bluetooth headset is a **PAN**; connecting computers inside a college lab is a **LAN**; connecting cable TV across a city is a **MAN**; and the **Internet** is the largest global **WAN**.

2. Network Topologies

Refers to the physical or logical layout of nodes and connections in a network.

- **Bus Topology:** All nodes share a single central cable (backbone).
- **Star Topology:** All nodes connect individually to a central hub or switch.
- **Ring Topology:** Devices are connected in a circular loop; data flows in one direction.
- **Mesh Topology:** Every device is connected to every other device (High redundancy).
- **Hybrid Topology:** Combination of two or more different topologies.

Real-World Example: Modern office LANs use **Star Topology** with Ethernet switches for easy troubleshooting and reliability. Critical data centers use **Full Mesh Topology** so that network failure never occurs.

3. Network Models: OSI vs TCP/IP

Standardized frameworks defining how network protocols interact during communication.

- **OSI 7-Layer Model:** Physical, Data Link, Network, Transport, Session, Presentation, Application.
- **TCP/IP 4-Layer Model:** Network Interface, Internet, Transport, Application.

Real-World Example: When you open a website (HTTP - Application Layer), TCP breaks data into segments (Transport Layer), IP assigns sender/receiver IP addresses (Network Layer), MAC addresses handle hop-to-hop transfer (Data Link Layer), and bits transmit over Fiber/Wi-Fi (Physical Layer).

4. IP Addressing & Subnetting

Unique logical addresses assigned to devices operating on a TCP/IP network.

- **IPv4:** 32-bit address represented in dotted-decimal format (e.g., 192 . 168 . 1 . 1).
- **IPv6:** 128-bit address in hexadecimal format to handle massive global device growth.

Real-World Example: Your home Wi-Fi router assigns a private IPv4 address like 192 . 168 . 1 . 5 to your laptop, while your Internet Service Provider (ISP) assigns a public IP to connect to web servers worldwide.

5. Essential Network Hardware Devices

- **Hub:** Layer 1 physical device that broadcasts incoming data to all connected ports.
- **Switch:** Layer 2 intelligent device that routes data frames directly to target MAC addresses.
- **Router:** Layer 3 device that routes data packets between different IP networks.
- **Gateway:** Connects networks using different protocols.

Real-World Example: A **Switch** connects computers within your office LAN, while a **Router** connects your office network to the external Internet.

3. Additional Important Concepts

- **Transmission Modes:** Simplex (One-way like TV broadcast), Half-Duplex (Two-way non-simultaneous like Walkie-Talkie), Full-Duplex (Simultaneous two-way like Telephone).
- **Transmission Media:** Guided (Twisted Pair, Coaxial Cable, Optical Fiber) and Unguided (Wi-Fi, Satellite, Microwave, Radio Waves).
- **Switching Techniques:** Circuit Switching (Dedicated link like PSTN) vs Packet Switching (Data broken into independent packets like Internet).

4. Computer Networks Practice Questions (25 MCQs)

1. How many layers are present in the OSI reference model?

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|------|------|
| A) 4 | B) 5 |
| C) 7 | D) 6 |

2. Which device operates at Layer 3 (Network Layer) of the OSI model?

- | | |
|-----------|-------------|
| A) Hub | B) Switch |
| C) Router | D) Repeater |

3. What is the size of an IPv4 address?

- | | |
|------------|-------------|
| A) 16 bits | B) 32 bits |
| C) 64 bits | D) 128 bits |

4. Which network topology requires a central hub or switch?

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|------------------|------------------|
| A) Bus Topology | B) Ring Topology |
| C) Star Topology | D) Mesh Topology |

5. HTTP operates at which layer of the OSI model?

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|----------------------|--------------------|
| A) Application Layer | B) Transport Layer |
| C) Network Layer | D) Data Link Layer |

6. What is the size of a MAC address?

- | | |
|------------|-------------|
| A) 32 bits | B) 48 bits |
| C) 64 bits | D) 128 bits |

7. Which protocol is reliable and connection-oriented?

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|--------|---------|
| A) UDP | B) TCP |
| C) IP | D) ICMP |

8. Walkie-Talkie communication is an example of which transmission mode?

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|----------------|----------------|
| A) Simplex | B) Half-Duplex |
| C) Full-Duplex | D) Multiplex |

9. Which topology offers maximum redundancy and fault tolerance?

- | | |
|------------------|------------------|
| A) Mesh Topology | B) Star Topology |
| C) Bus Topology | D) Ring Topology |

10. What is the size of an IPv6 address?

- | | |
|-------------|-------------|
| A) 32 bits | B) 64 bits |
| C) 128 bits | D) 256 bits |

11. ARP (Address Resolution Protocol) maps:

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|------------------------------|------------------------------|
| A) MAC address to IP address | B) IP address to MAC address |
| C) Domain name to IP address | D) Port number to IP address |

12. DNS stands for:

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|---------------------------|---------------------------|
| A) Data Network System | B) Domain Name System |
| C) Digital Network Server | D) Domain Network Service |

13. Which port is used by default for HTTP?

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|-------|--------|
| A) 21 | B) 25 |
| C) 80 | D) 443 |

14. Which port is used by default for secure HTTPS communication?

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|--------|-------|
| A) 80 | B) 22 |
| C) 443 | D) 53 |

15. What type of physical media transmits data using pulses of light?

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|------------------------|--------------------------|
| A) Coaxial Cable | B) Twisted Pair Cable |
| C) Optical Fiber Cable | D) Shielded Twisted Pair |

16. Which collision resolution protocol is used in Ethernet networks?

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|------------|------------------|
| A) CSMA/CD | B) CSMA/CA |
| C) ALOHA | D) Token Passing |

17. Which protocol automatically assigns IP addresses to devices on a network?

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|--------|---------|
| A) DNS | B) DHCP |
| C) FTP | D) SMTP |

18. What is the standard loopback IPv4 address?

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|----------------|--------------------|
| A) 192.168.1.1 | B) 10.0.0.1 |
| C) 127.0.0.1 | D) 255.255.255.255 |

19. FTP (File Transfer Protocol) uses which pair of port numbers?

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|---------------|----------------|
| A) 20 and 21 | B) 22 and 23 |
| C) 25 and 110 | D) 80 and 8080 |

20. Which protocol is connectionless and lightweight for fast data streaming?

- A) TCP
- B) UDP
- C) HTTP
- D) SSH

21. Network layer data packets are framed at which layer?

- A) Physical Layer
- B) Data Link Layer
- C) Transport Layer
- D) Presentation Layer

22. What does NAT stand for?

- A) Network Address Translation
- B) Network Access Terminal
- C) Native Address Transfer
- D) Network Protocol Topology

23. Which command utility tests network connectivity to a remote IP address?

- A) ipconfig
- B) ping
- C) netstat
- D) nslookup

24. What is the primary function of a Firewall?

- A) Increase network bandwidth
- B) Filter network traffic for security
- C) Assign IP addresses automatically
- D) Boost Wi-Fi signals

25. Data unit at the Transport Layer of the OSI model is called:

- A) Bits
- B) Frames
- C) Packets
- D) Segments

Answer Key

- | | | | | |
|------|-------|-------|-------|-------|
| 1. C | 6. B | 11. B | 16. A | 21. B |
| 2. C | 7. B | 12. B | 17. B | 22. A |
| 3. B | 8. B | 13. C | 18. C | 23. B |
| 4. C | 9. A | 14. C | 19. A | 24. B |
| 5. A | 10. C | 15. C | 20. B | 25. D |